Martin Wu

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Education

University of British Columbia **Expected May 2029**

Bachelor of Applied Science in Computer Engineering *Internship availability: Jan or May 2027, 4, 8 or 12 months*

- Average: A- | Coursework: Computer Systems (SystemVerilog, FPGA, RISC-V), Software Construction (Java)

Experience

Embedded Systems Developer **Sep 2025 – Present**

UBC Solar *Vancouver, BC*

- Brought up the Motor Drive Interface firmware on STM32: pin setup, a CAN driver with a hardware filter for motor commands, and ADC readings of board voltage and temperature
- Made motor commands expire after 100 ms of CAN silence, so a stale command cannot keep driving the motor; confirmed on hardware by cutting the command stream
- Proposed PT1000 RTD motor-temperature sensing with a MAX31865 and wrote its firmware from the first prototype to the SPI driver; verified with heat-gun tests and integration-tested with the car's electrical system
- Built RTD fault handling: a leaky-counter debounce latches after 5 bad reads, fault and SPI-error bits go out on CAN, and a config read-back catches a missing chip that would otherwise read −259 °C
- Wrote external-lights firmware as FreeRTOS tasks (turn signals, hazards, brake lights) and steering-wheel code that shows CAN speed data in km/h or mph on the hex display
- Added the new CAN signals to the telemetry DBC, the Grafana dashboards and the pit-crew test, so the crew sees motor-board data and sensor faults live

Co-Founder & Software Engineer **Oct 2025 – Present**

Stealth AI Infrastructure Startup *Vancouver, BC*

- Wrote a Go test harness whose in-memory adapters drive reads, idempotent replays, identity denials and adapter panics through the team's 11-step dispatcher without Docker
- Fixed the tooling and rewrote the demo script for the live product demo, where AI agents query Postgres through access policies, human approvals and an audit trail; it is used in investor meetings and accelerator applications
- Ran the technical review for the first enterprise deal, an on-prem AI cluster now at the LOI stage: showed the model would not fit unquantized, flagged support and purchasing risks, and drafted the LOI

Projects

Songless | *React, TypeScript, Vite, Cloudflare R2* | Live **2026**

- Built a music-guessing web game with **100k+ visits** and 804 songs; 30-second clues stream from R2, and the full track loads only on the results screen
- Wrote the upload pipeline: trims leading silence, encodes 128 kbps MP3s, and refuses any batch that would push the bucket past an 8.5 GB ceiling

Useful Tool Hub | *JavaScript, Python, Jest, Vercel* | Live **2026**

- Made a dozen web tools (file conversion, OCR, transcription, PDF signing) that run in the browser, with a Python backend only for downloaders; Jest runs in CI and scripts check the deployed site

Us (browser story game) | *Phaser 4, TypeScript, Playwright* | Live **2026**

- Created 15+ scenes in Phaser 4 with saves, fast travel and dialogue rewind; Playwright scripts play through routes and check lighting and collision

Skills & Awards

Languages: C, Python, Go, TypeScript, JavaScript, Java, SystemVerilog
Embedded: STM32 HAL, FreeRTOS, CAN bus, SPI, ADC, CMake, oscilloscope, PCAN, J-Link, ST-Link
Frameworks & Tools: Linux, Git, React, Vite, Docker, Grafana, Playwright, Jest
Awards: CSMC Top 0.5% nationally (71st) | AIME Qualifier 2x | Euclid Top 2% (85/100)